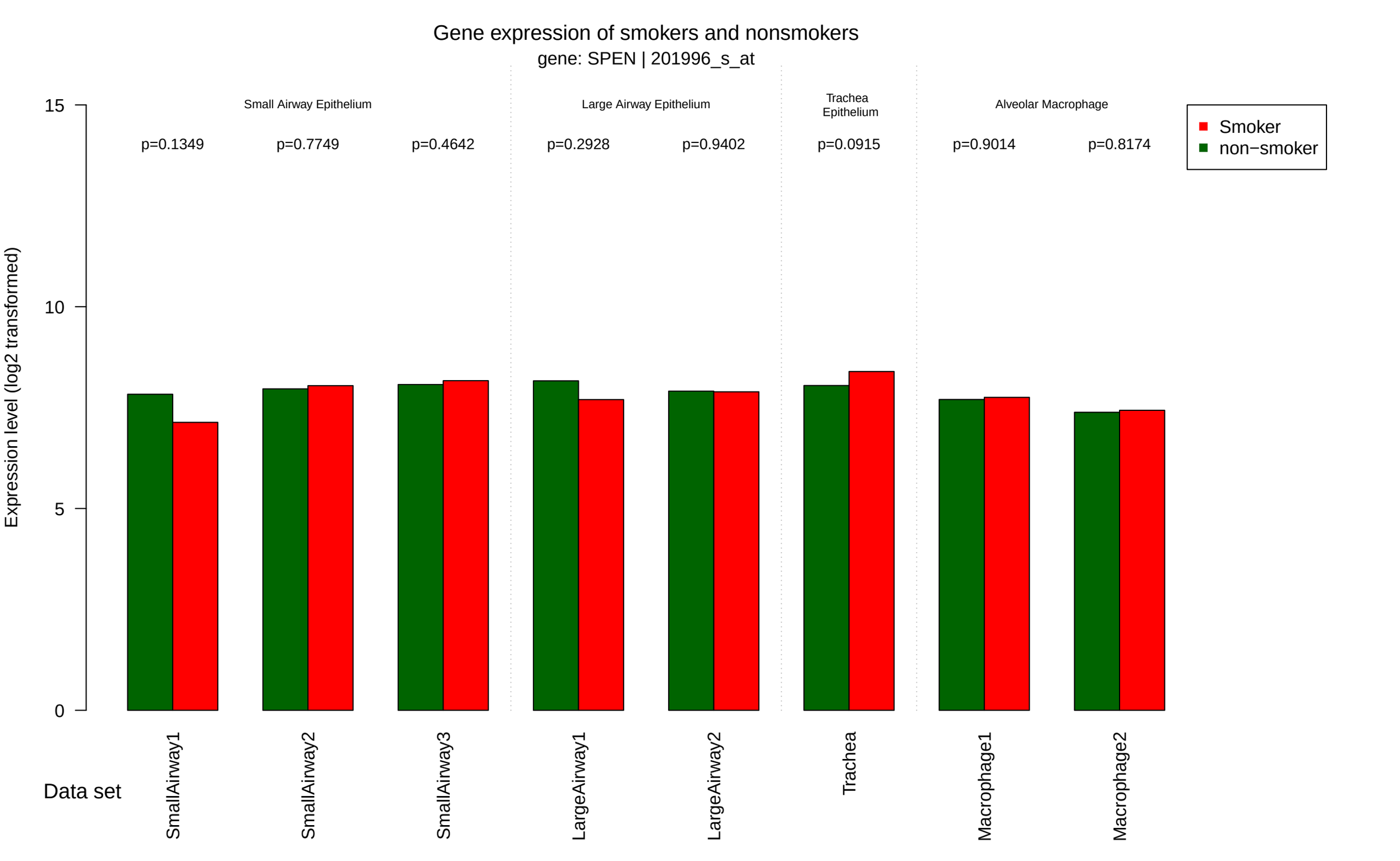
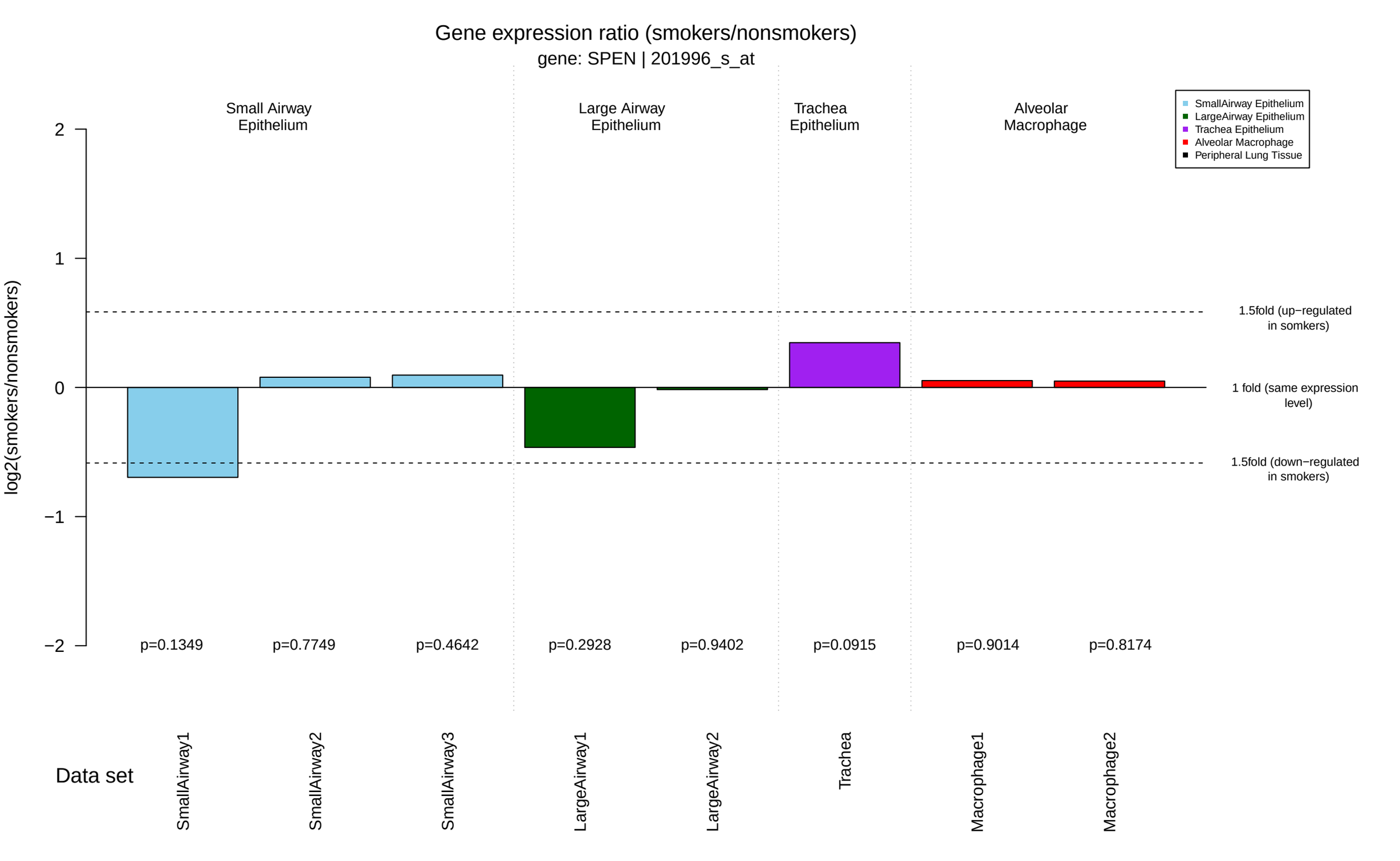
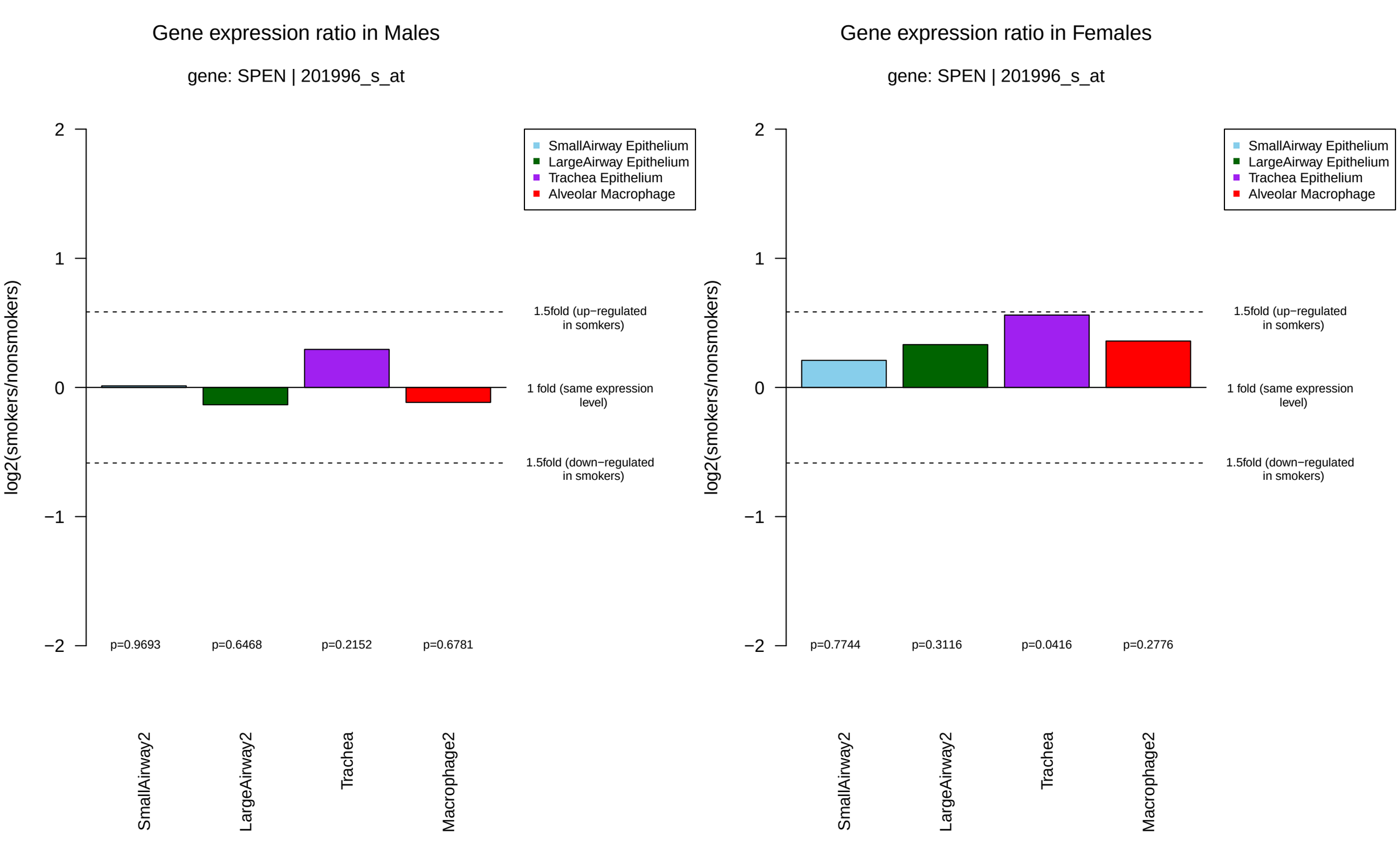
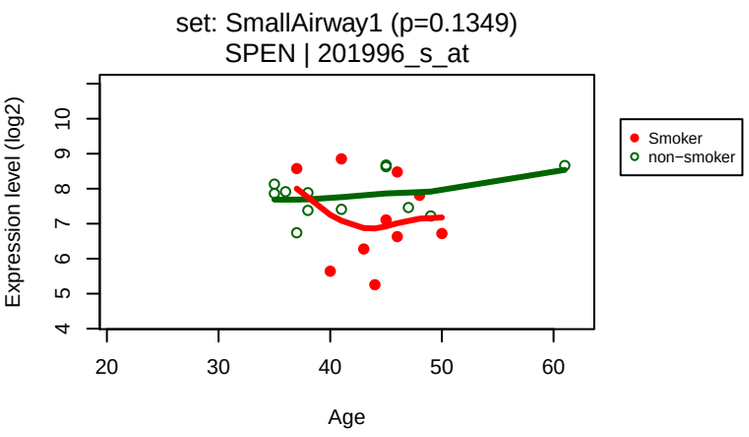
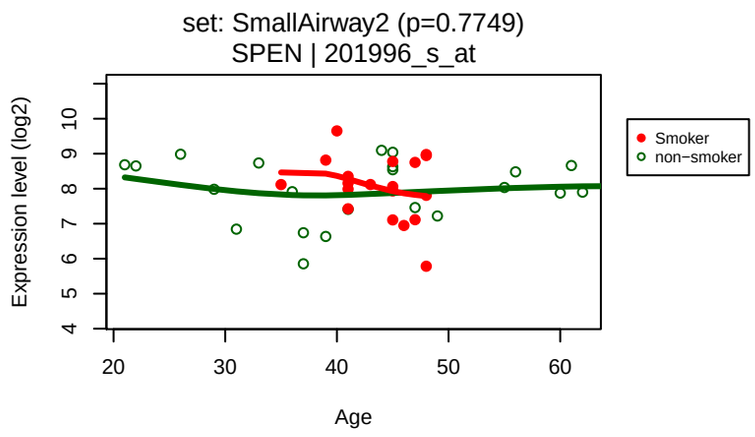
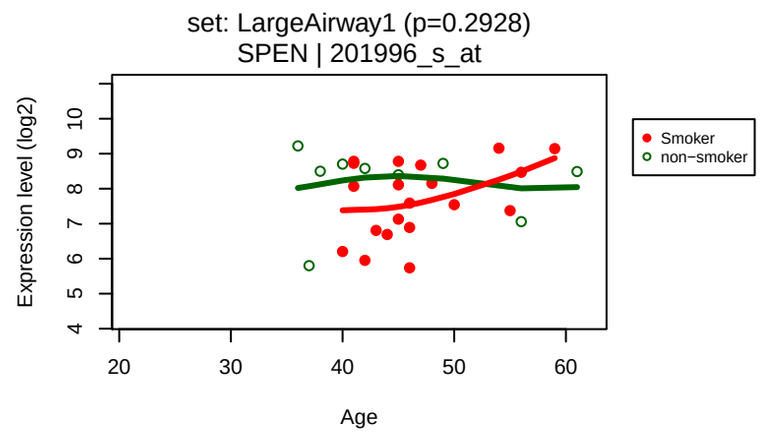

Gene expression of smokers and nonsmokers
gene: SPEN | 201996_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1349
p=0.7749
p=0.4642
p=0.2928
p=0.9402
p=0.0915
p=0.9014
p=0.8174
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

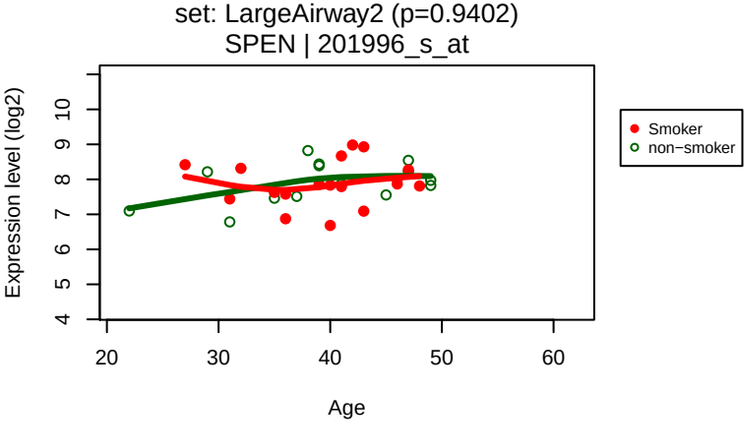

Gene expression ratio (smokers/nonsmokers)
gene: SPEN | 201996_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1349
p=0.7749
p=0.4642
p=0.2928
p=0.9402
p=0.0915
p=0.9014
p=0.8174
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

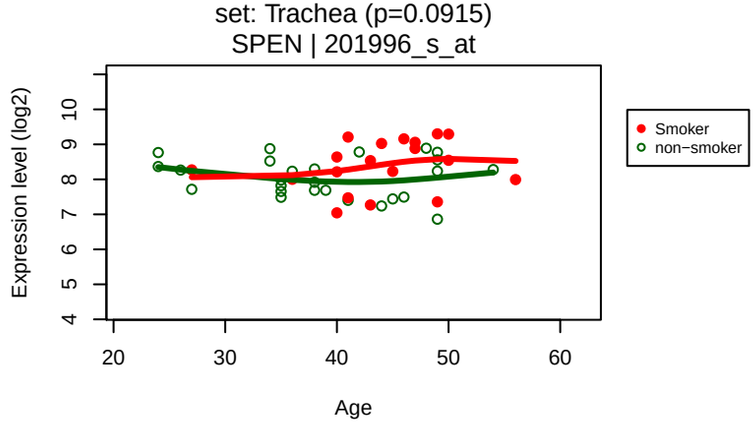

Gene expression ratio in Males
gene: SPEN | 201996_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9693
p=0.6468
p=0.2152
p=0.6781
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPEN | 201996_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7744
p=0.3116
p=0.0416
p=0.2776
SmallAirway2
LargeAirway2
Trachea
Macrophage2

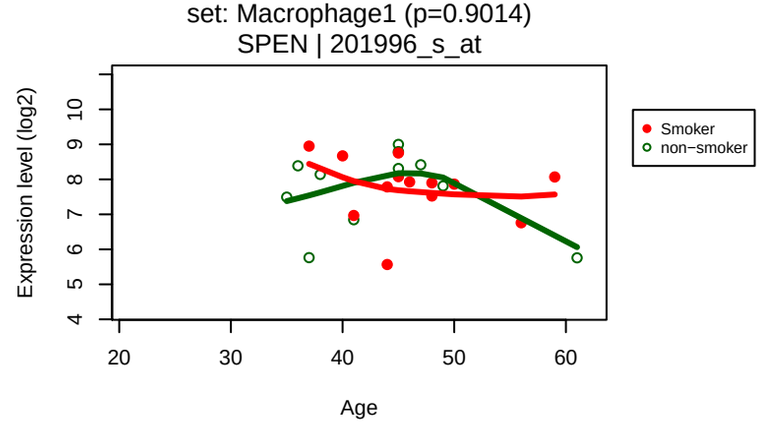

set: SmallAirway1 (p=0.1349)
SPEN | 201996_s_at
Expression level (log2)
Age
Smoker
non−smoker

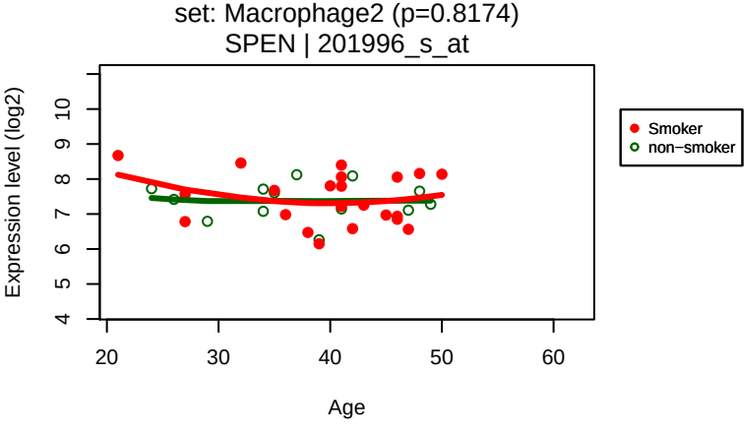

set: SmallAirway2 (p=0.7749)
SPEN | 201996_s_at
Expression level (log2)
Age
Smoker
non−smoker

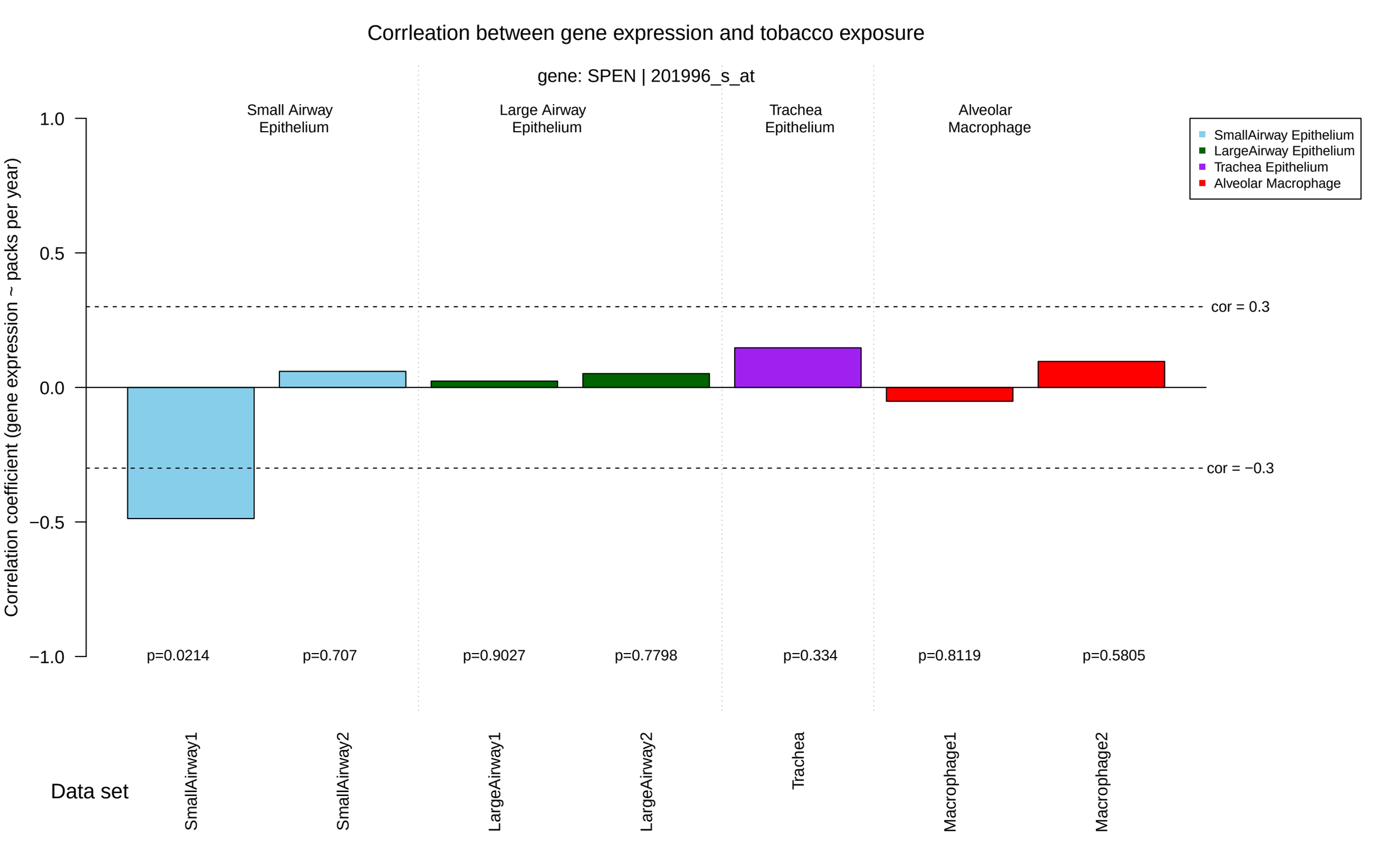

set: LargeAirway1 (p=0.2928)
SPEN | 201996_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9402)
SPEN | 201996_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0915)
SPEN | 201996_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9014)
SPEN | 201996_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8174)
SPEN | 201996_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPEN | 201996_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0214
p=0.707
p=0.9027
p=0.7798
p=0.334
p=0.8119
p=0.5805
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2